Fine-scale modeling for crown-of-thorns starfish control on the Great Barrier Reef: directing daily search effort and testing strategic interventions

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Abstract

Reef interventions are moving from field trials to program-scale operations, and their efficiency now depends on decision support that works at the scales where crews and managers act. Control of crown-of-thorns starfish (COTS) on the Great Barrier Reef is a program-scale reef intervention in which two such decisions recur: where on a reef to search and cull each day, and which reefs to treat and when to start or stop control. We developed fine-scale models for both. For daily operations, we ranked all 15,884 designated cull sites on 576 reefs by combining 30 m habitat suitability models with the previous season's cull dives and manta tow surveys, and tested it within reefs on held-out data against a no-information baseline at equal effort. Over 2018–2026, the combined ranking reduced the share of designated sites searched from 88% to 82% while retaining 90% of dives that found problematic COTS densities (at least 0.02 COTS per minute of bottom time). It also avoided 6 and 8 percentage points more low-density and empty dives than habitat alone. Fitted only on earlier seasons, it outperformed habitat alone in every season from 2022. On its own, habitat ranked locations within reefs only modestly; its value lay in ordering search where recent evidence was missing. For manta surveillance, evenly spaced tows retained at least 95% of aggregations with 40% fewer tows. For strategic decisions, we developed a site-resolved metacommunity model, COTSMod.jl coupled to the ADRIA reef ecosystem model,

for 2,895 sub-reef sites on 113 reefs around Lizard Island. It links COTS demography, coral predation and larval connectivity with prescribed historical culling. 3 candidates matched all 7 historical peaks across 5 seed pairs, but recent build-up and validation remain unresolved. Comparisons of primary-outbreak suppression and starting and stopping rules remain future work. The ranking has been delivered for the 2027 control season. The ranking directs daily search; the metacommunity prototype provides a platform for testing strategic control.

Keywords: crown-of-thorns starfish, decision support, reef interventions, species distribution model, metacommunity model, outbreak control, scaling strategies, Great Barrier Reef

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Introduction

Coral reefs are declining under cumulative pressures, and keeping them functional increasingly depends on active interventions, from pest control and restoration to spatial management, delivered at scales far beyond the plots where they were first tested (Anthony et al., 2017). Scaling an intervention changes the problem it has to solve. In a trial, the question is whether the intervention works where it is applied. At program scale, much of the cost lies in deciding where and when to apply it, and those decisions are made by crews and managers at scales of tens of meters to tens of kilometers. Decision support can sharpen those choices, but only if it is built at the scale of the decision and judged against what operators would achieve without it.

Control of crown-of-thorns starfish (*Acanthaster* cf. *solaris*; COTS) on the Great Barrier Reef (GBR) is one of the few reef interventions already operating at that scale. Outbreaks of COTS have been a major cause of coral loss on the GBR (De'ath et al., 2012; Pratchett et al., 2017), and the COTS Control Program suppresses them by manual culling, which reduces COTS densities and protects coral when it is sustained at the right reefs (Babcock et al., 2020; Westcott et al., 2020). Most of the program's vessel and diver time goes into finding COTS. Manta tow surveys, in which an observer is towed around a reef, usually along its perimeter, locate COTS and establish a reef's outbreak status. Cull crews then dive designated sites, chosen largely from manta results and from where COTS were found before. Much of this effort finds little. Of the held-out cull dives analyzed here, 18% found no COTS and a further 21% found them below the density at which culling is considered effective.

Two kinds of decision determine how efficiently this effort is used, and they operate at different fine scales. Day to day, crews decide where on a reef to tow and which designated sites to dive. Their best information is what they found last season, which is rich on reefs they visit often and thin on reefs and parts of reefs they rarely search. Strategically, managers decide which reefs to treat and when. COTS larvae disperse among reefs, so an outbreak on one reef can seed others downstream, and the spread of outbreaks depends on the structure of the connectivity network (Hock et al., 2014, 2017). Outbreaks on the GBR have tended to begin on reefs in a northern initiation region and spread south, which has led to proposals to concentrate control there early in an outbreak cycle (Pratchett et al., 2017; Babcock et al., 2020). The same reasoning raises operational questions that the program faces every season: when to begin culling on a reef as densities rise, when to stop on a heavily affected reef, and what either choice means for the reefs

it supplies with larvae.

Modeling has so far addressed these decisions at the scale of whole reefs. Reef-level distribution models have shown that larval connectivity and water quality explain much of where outbreaks occur across the GBR (Matthews et al., 2020a), and hybrid machine-learning models now forecast which reefs are likely to be in outbreak in a given year (Stewart et al., 2026). Models of intermediate complexity and reef-scale ecosystem models, including MICE, CoCoNet and ReefMod-GBR, represent COTS and coral dynamics and have been used to compare control strategies across the reef system (Morello et al., 2014; Plagányi et al., 2014; Condie et al., 2021; Bozec et al., 2022). Because they treat each reef as a single unit, they cannot represent where on a reef COTS aggregate, where crews search, or how culling at particular sites propagates through connected reefs. Two developments now make a finer scale feasible. Coral species distribution models built from multiscale terrain variables map habitat for key coral taxa at tens of meters (Guillaume et al., 2025), and COTSMod, a spatially explicit metacommunity model of COTS outbreaks and coral recovery (Matthews et al., 2020b), showed how outbreaks can be simulated across connected sites. Reef ecosystem frameworks such as ADRIA, with the computationally efficient CoralBlox coral model, can now run the large numbers of scenarios that comparisons of control strategies require (de Almeida et al., 2026; open-AIMS, 2026b).

Working at fine scales raises its own problem of evaluation. Fine-scale species distribution models (SDMs) can map COTS habitat at tens of meters (Elith and Leathwick, 2009), but COTS control data make them hard to evaluate. Observations are preferentially sampled, because crews dive where manta tows or past experience suggest COTS are present, so a model fitted to them partly learns operator behavior and in-sample scores overstate its skill (Phillips et al., 2009; Diggle et al., 2010). Survey year and reef identity dominate raw probabilities, so pooled discrimination metrics reward separating outbreak reefs from quiet ones instead of ranking places within a reef (Lobo et al., 2008). Credible evaluation therefore requires held-out data at the unit of the decision (Roberts et al., 2017) and a baseline with the same effort.

Here we develop fine-scale models for both kinds of decision in GBR COTS control. For day-to-day operations, we ask whether habitat suitability models, combined with the previous season's cull and manta evidence, can reduce manta tows and cull-site searches without missing COTS aggregations or problematic densities. We test the search rankings on held-out reefs and forward in time against no-information baselines, and deliver a priority ranking of every designated cull

site for the 2027 season. For strategic decisions, we extend COTSMod into a site-resolved metacommunity model of COTS and coral, coupled to ADRIA and CoralBlox, for the Lizard Island cluster within the northern initiation region. We assess how well it reproduces observed outbreak cycles and recorded historical removals, as prerequisites for future comparisons of primary-outbreak suppression and starting and stopping rules, including their consequences for connected reefs downstream.

Materials and methods

Model reporting follows the ODMAP structure for species distribution models where it applies (Zurell et al., 2020).

Data and response definitions

Cull dives from the COTS Control Program provided the site, date, number of COTS culled and bottom time for each dive. Records with missing counts or non-positive bottom time were excluded. Dives were matched to the November 2025 EoTR designated cull-site register by reef and standardized site name, with a spatial fallback to the nearest polygon within 100 m on the same reef. This matched 93.8% of dives. The register contained 15,884 site polygons on 576 reefs; 4,299 sites had been dived. [TO CONFIRM: years of cull records used for the archived habitat fits and the data-access agreement.]

A problematic dive had catch per unit effort (CPUE) ≥ 0.02 COTS per minute of bottom time. Dives that found COTS below this threshold were low-density dives; those that found none were ghost dives. Habitat models used the archived problematic-density indicator at the cull site-year scale, while operational selection was evaluated against individual dive outcomes.

The manta-trained model used 198,852 labeled tows in 1,015 reef-years, including 16,199 with strong COTS evidence. Strong evidence meant a direct COTS sighting, at least two counted feeding scars, or scars recorded as “common” in legacy surveys. Strict negatives recorded neither COTS nor scars; weak single-scar records were excluded from model fitting. Cull and manta records were collected by program crews and held by GBRMPA. [TO CONFIRM: years covered by the manta records.] SALAD imagery surveys and Reef Health and Impact Surveys (RHIS) contributed to the explanatory joint model only. [TO CONFIRM: SALAD and RHIS

custodians, sampling methods and years.]

Predictors were assembled on a 30 m grid in GDA2020 / MGA zone 55. The clean stack contained depth, slope, eastness, northness, bathymetric position index, vector ruggedness, sky-view factor, horizontal and vertical curvature, and three coral habitat-suitability surfaces for *Acropora* taxa (Guillaume et al., 2025). Five ReefGuide layers added significant wave height, wave period, turbidity, bathymetry and slope. The joint model also used geomorphic classes and reef features. [CITATION NEEDED: bathymetry, ReefGuide and GBR10 geomorphic-map sources.] [TO CONFIRM: that AhyaD, Aspat and Aten are the Guillaume et al. (2025) models, and predictor dates and resampling methods.]

Habitat models and mapped layers

The clean + year model used gradient-boosted trees [XGBoost; Chen and Guestrin (2016)] to predict the problematic-density indicator from the clean stack and survey year. The RG + year model added the five ReefGuide predictors. These cull-trained models learned from locations selected by crews, often following manta evidence, so their scores reflect the program's sampling footprint as well as habitat. The operational refits used 1,500 boosting rounds, a learning rate of 0.00137382379588326, maximum tree depth 5, minimum child weight 20 and gamma 10^{-10}, retaining the archived hyperparameters. These settings were not retuned on the evaluation reefs.

The primary search-planning layer used RG + year where available and clean + year elsewhere, calibrated onto the RG probability scale before combining them. Each 30 m cell was then ranked within its reef polygon, with map priority classes for the top 10%, 10–20% and 20–30%. RG + year covered 61% of the reef-polygon support cells used in the coverage analysis. Of the mapped reefs, 54 used RG alone, 2,586 mixed the two sources and 920 used the clean fallback alone.

Manta-trained XGBoost models used the RG stack and the same boosting settings. Predictor values were averaged along each tow. A model with survey year was compared with a model designed specifically to rank locations within reef-years. For the latter, the logit of the reef-year's strong-tow proportion, with a half-count correction, entered training as a fixed offset and survey year was omitted. The offset was omitted when predicting relative habitat scores because it was constant within a reef-year. The resulting map represents relative log-odds, not an annual occurrence probability. The cull-trained and manta-trained ranks were also compared

as an average rank and as separate terms in the evidence model.

The hierarchical model of species communities [HMSC; Ovaskainen et al. (2017); Tikhonov et al. (2020)] related cull, SALAD and RHIS responses jointly to habitat at 120 m, conditional on an active reef-year. Three predictor components supplied fallbacks where enhanced predictors were unavailable. Production fits used four chains, each retaining 1,000 samples at thinning 30 after 5,000 transient iterations. Convergence was assessed against a minimum effective sample size of 400 and maximum upper potential scale reduction factor of 1.1. This model supplied explanatory habitat effects and uncertainty.

An additional layer represented known aggregation sites using a Gaussian kernel of the problematic rate in earlier active cull site-years ($\sigma = 500$ m). Its rate H and evidence weight W updated the HMSC habitat probability p as $(\kappa p + WH)/(\kappa + W)$, with $\kappa = 2$ for the fixed-setting comparison. History and dive effort were interpreted as records of past finds and crew targeting, rather than independent habitat evidence.

Evaluation design and search-efficiency analyses

Operational discrimination was measured among locations on the same reef in the same year. Within reef-year AUC compared positive and negative locations only within those groups; an AUC of 0.5 indicated no discrimination. Top-20% lift was the share of positives in the highest-ranked fifth divided by 0.2. These metrics were supplemented by the outcomes retained when effort was reduced.

For operational habitat comparisons, reefs were assigned to five folds and every reef was scored by a model refitted without any reefs in its fold (Roberts et al., 2017). We refer to this as leave-reef-out validation; it is grouped five-fold validation, rather than one refit per reef. This prevented a reef's cull or manta observations from contributing to its habitat fit. A separate matched manta check scored the same 117,318 tows in 1,003 active reef-years on 402 reefs where all deployed layers were available. That check used the fitted layers and was not a substitute for the refitted operational evaluation.

Manta aggregations were connected groups of strong-evidence tow midpoints linked within 500 m in a survey (reef and date); successive links could join a group wider than 500 m. The principal aggregation analysis required at least two strong tows, with isolated records examined separately. Surveys required at least five scored tows and some strong evidence. An aggre-

gation was retained if at least one of its tows was kept. This measured retention of observed aggregation evidence under retrospective thinning.

Each thinning strategy kept the same number of tows per survey, over retained shares from 90% to 30%. Model thinning kept the highest-scoring tows. Random thinning used the exact hypergeometric expectation of retaining at least one tow in each aggregation. Systematic thinning kept evenly spaced positions in the tow sequence, averaged over 20 starting offsets. Recorded tow order was unavailable, so sequence was approximated by bearing from the survey center. The comparison therefore concerned thinning existing surveillance, not the performance of newly designed tow routes.

Cull-site analyses ranked every designated site on a reef, including never-dived sites, by its mean habitat score over the polygon. For each retained share of designated sites, the analysis recorded the share of observed problematic, low-density and ghost dives at retained sites. Random selection supplied a no-information expectation: retaining a fraction of sites gives that fraction's expected retention of each observed outcome. Outcomes at never-dived sites were unknown, so this exercise could not measure COTS missed at those sites or changes in crew behaviour under deployment.

To add recent evidence, each site received indicators for any problematic dive there in the previous season and for having been dived with none problematic. Neighborhood terms were the problematic rate and dive effort among nearby cull dives, and strong-evidence and total manta tows, using a Gaussian kernel with $\sigma = 300$ m. Effort and tow counts entered logistic regressions after $\log(1 + x)$ transformation; missing habitat ranks were assigned the midpoint. Habitat alone, evidence alone, habitat + evidence and a simple flagged-sites-first rule were compared. The regressions were fitted on other reef folds, and evidence for a test year came only from preceding seasons. This spatial test did not also restrict regression training outcomes to years before the test year.

Rankings were compared at matched retention of problematic dives because evidence could move both problematic and low-density dives upwards. Retention curves were evaluated over 5% steps in designated sites searched and interpolated at 80%, 85%, 90% and 95% problematic-dive retention. The separate season-by-season test used 80% and 90%. For each test year from 2019 to 2026, its evidence regression was fitted only on earlier dive outcomes, comparing all earlier seasons with the last five or three, and cull-only, averaged cull–manta or separate habitat

terms. Habitat covariates in this test were the reef-held-out ranks from the spatial analysis; the habitat fits themselves were not restricted to preceding years. Thus the evidence regression was tested forward in time, while the habitat component was held out spatially.

The HMSC–history forecast used a separate temporal design: for each test year from 2022 to 2026, habitat models were refitted on earlier years and history used earlier active site-years only. First-visit cull site-years were compared within reef-years containing both outcomes, using habitat, prior dive effort, history and their blend. This distinguished habitat transfer from the persistence of previous finds.

Uncertainty intervals used paired bootstrap resamples of reef-years for tow and cull metrics, or surveys for aggregation retention. Operational spatial comparisons used 500 resamples; the season-by-season site-ranking test used 200. The same resampled units were used across scores to preserve paired comparisons.

The evaluation design is summarized in Table 1.

Table 1: Evaluation design. Each question is evaluated on the unit and with the hold-out at which the decision is made, against a baseline at the same effort.

Question	Evaluation unit	Hold-out	Baseline
Within-reef ranking skill	Manta tows within reef-years	Reef folds for refits; separate deployed-layer check	AUC 0.5; lift 1
Manta thinning	Aggregations within surveys	Reef folds	Random and systematic thinning
Cull-site selection	Cull dives within reef-years, all designated sites	Reef folds	Random (retention = share of dives kept)
Adding last season's evidence	As for cull-site selection	Reef folds; evidence from season T−1 only	Habitat-only ranking at matched retention
Persistence over time	Cull dives within each test season	Evidence regression: seasons < T; habitat: reef folds	Habitat-only ranking

Seasonal products

For 2027, the selected cull-habitat + evidence model was fitted on 28,431 dives from 2019-2026. Recent evidence covered 1 June 2025 to 31 May 2026, the latest 12 months available, because calendar 2026 contained only January–May. Sites were ranked within reefs and assigned classes A (top 20%), B (20–50%), C (50–80%) and D (bottom 20%). The deployed habitat layer supplied scores for this product; the operational validation used refitted habitat scores.

Exploration areas were drawn from 30 m cells 0.5–16 m deep in the reef's top 30% of habitat, more than 100 m from designated cull sites and more than 150 m from manta tows in the preceding five seasons. Habitat was the mean of the primary-layer and manta-model percentiles, reranked within the reef, with the primary layer alone where manta scores were missing. Nearby pockets were joined, large patches split into compact areas of about 9 ha, and outlines smoothed and clipped to the reef outside cull sites. Areas smaller than 0.5 ha were removed; up to ten areas per reef polygon, at least 350 m apart, were retained. These areas identified places to explore beyond the recent search footprint.

Metacommunity model and historical culling

The strategic modeling strand extended COTSMod (Matthews et al., 2020b) through the Julia package COTSMod.jl, coupled to ADRIA and the CoralBlox coral model (de Almeida et al., 2026; open-AIMS, 2026c, 2026a). COTSMod.jl supplied population transitions, coral predation and larval processes; ADRIA supplied the spatial domain, disturbance forcing and scenario execution. The prototype used the Lizard Island cluster, with 2,895 sub-reef sites on 113 reefs. Historical simulations beginning in 1991 were assessed against AIMS Long-Term Monitoring Program (LTMP) manta COTS and coral records on Lizard Island, Eyrie, MacGillivray and North Direction reefs. The model state was expressed as recruit, juvenile and adult densities per hectare of site area, with coral cover represented relative to habitable area.

The working COTS formulation combined preferred-first per-capita grazing, survival linked to remembered preferred coral cover, adult senescence and preferred-food effects on reproductive condition. Larval production included a low-density Allee effect, juvenile survival included a saturating predator-pit term, and larvae were transported using reef-level connectivity and supplied from outside the domain. External supply was derived from the reef-level outbreak-probability hindcast (Stewart et al., 2026); its production multiplier was a calibration assumption,

not measured recruitment. Mechanisms suggested by MICE, CoCoNet and ReefMod-GBR were examined as optional alternatives (Morello et al., 2014; Plagányi et al., 2014; Condie et al., 2021; Bozec et al., 2022). Testing a mechanism did not make it part of the working configuration. [TO CONFIRM: attribution and final choice of the annual connectivity matrices; the sub-reef hydrodynamic network had not replaced these reef-level inputs.]

Calibration used bounded comparisons of candidate mechanisms and parameter settings. Each screen was preregistered, included an exact-replay reference and recorded settings, seeds, input checksums and repository revisions. Outbreak peak presence, timing and height, inter-outbreak troughs and coral trajectories were assessed together. Coral dynamics were also assessed with COTS grazing forced from observed records to separate coral-model errors from population-dynamics errors. Simulated adult density was converted to manta CPUE through an explicit observation model. The working linear conversion used $CPUE = qA$, with A in adults ha^{-1} and q = 0.015 COTS tow^{-1} per adult ha^{-1}; alternative scales and density-dependent detection, $CPUE = qA^2/(A + A½)$, were sensitivity analyses. These alternatives were kept distinct because they change both initial-density estimates and the comparison with observations. In-sample calibration and diagnostic screens were not treated as held-out validation or evidence of management benefit.

The replicated historical comparison used score v2, which scored North Direction against its raw series and omitted Eyrie's unobserved first wave from the peak count. Its three retained candidates were compared over five recorded seed pairs. The later v0.4 diagnostic extended forcing to 2026 and used score v3: the manta-based score on 1992–2021 plus 1.5 times a diver-anchored recent-density log error. Original and cull/SALAD-corrected outbreak boundaries were crossed with earlier and revised coral thermal settings; exploratory chlorophyll-derived larval-survival modifiers were also screened. These later runs used one seed pair and prescribed culling was off. Their corrected boundary and recent-density score shared evidence, so the comparison was a diagnostic fit rather than a validation. The two score versions were reported separately.

Observed historical culling was included as prescribed removals. Cull records from the licensed CSIRO compilation were retained for reefs in the domain, grouped by site and year, and mapped to model sites on the same reef. Counts were distributed with weights proportional to site area $\times \exp(-d^2/(2\sigma^2))$ within 3σ of the cull-site centroid, normalized to preserve total counts. The primary width was σ = 210 m, with nearest-site allocation (σ = 0) and σ = 420 m as sensitivities;

the nearest site was used if no centroid lay within the kernel. Allocated counts were divided by model-site area to obtain removals in COTS ha^{-1}. COTS below 15 cm were assigned to the juvenile stage and those at least 15 cm to adults, an explicit size-to-stage approximation.

September–December removals were assigned to the following LTMP reporting year and applied before spawning; January–August removals were applied after spawning in their reporting year. The annual implementation placed these removals immediately before or after the COTS predation and population-transition call. Requested removals were capped at the simulated juvenile or adult population present, recruits were left unchanged, and adult age and size substates were scaled by the surviving adult fraction. The model logged actual removals separately from requested removals, allowing a shortage of simulated COTS to be diagnosed rather than removing animals that were absent.

This culling implementation prescribed observed catches. It did not yet predict diver search, catchability, effort costs or adaptive starting and stopping decisions. The intended scenarios—primary outbreak suppression, starting and stopping rules, and downstream consequences—require further calibration and independent validation before their management performance can be assessed.

Software and reproducibility

The SDM analyses used R 4.5.0 with xgboost, Hmsc and terra. Inputs were pinned by SHA-256 checksum in the final pipeline configuration, and reported results were read from output tables when the documents were rendered. [TO CONFIRM: package versions for the archived fits and a release identifier for the final analysis.] Metacommunity methods were checked against local COTSMod.jl and ADRIA code on 15 October 2026. The citations record both repository revisions and declare the uncommitted ADRIA changes inspected for this pass. [TO CONFIRM: Julia and package versions and the frozen code release to accompany submission.]

Results

Habitat models rank locations within a reef only weakly

Across the GBR, the absolute values of the habitat layers mostly reflected survey year and region (Figure 1). Within reefs, the cull-trained layers ranked reef margins and outer slopes highest

(Figure 2). On independent manta tows, the best cull-trained layer (RG + year) reached a within reef-year AUC of 0.576 (95% CI 0.548–0.602) (Table 2). Clean + year was close to chance at 0.516. The manta model ranked held-out tows best, at 0.612 (95% CI 0.578–0.636), with a top-20% lift of 1.33. Its strongest predictors were wave period and height, bathymetry and slope.

The layers also disagreed on which cells rank highest. Median within-reef Spearman correlations ranged from 0.24 to 0.41 between pairs of layers. Where both cull-trained layers existed, their rankings correlated at 0.34.

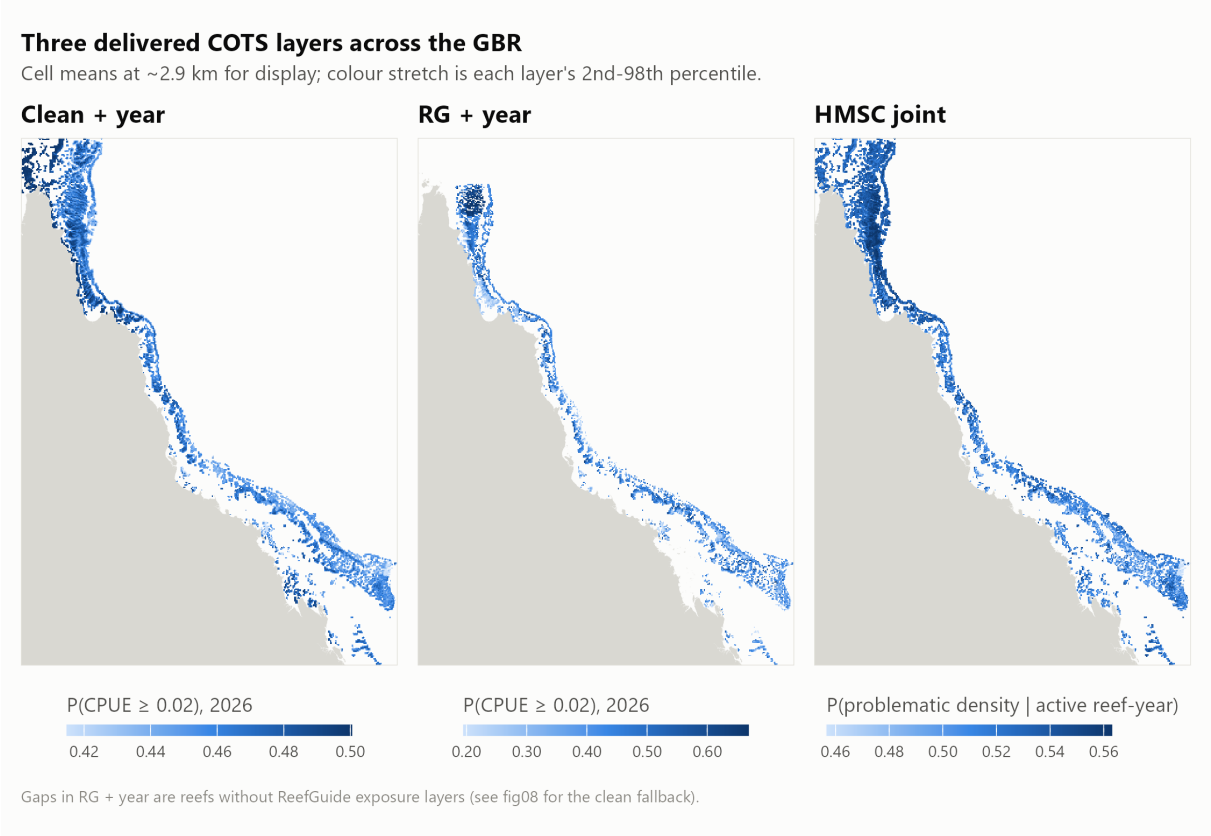


Figure 1: Habitat layers across the Great Barrier Reef, shown as cell means at about 2.9 km. The two cull-trained layers (left, center) are annual probabilities of a problematic cull site-year in 2026; the joint model (right) is conditional on an active reef-year. Gaps in RG + year are reefs without ReefGuide layers. At this scale the layers mainly show regional differences, not the within-reef structure used operationally.

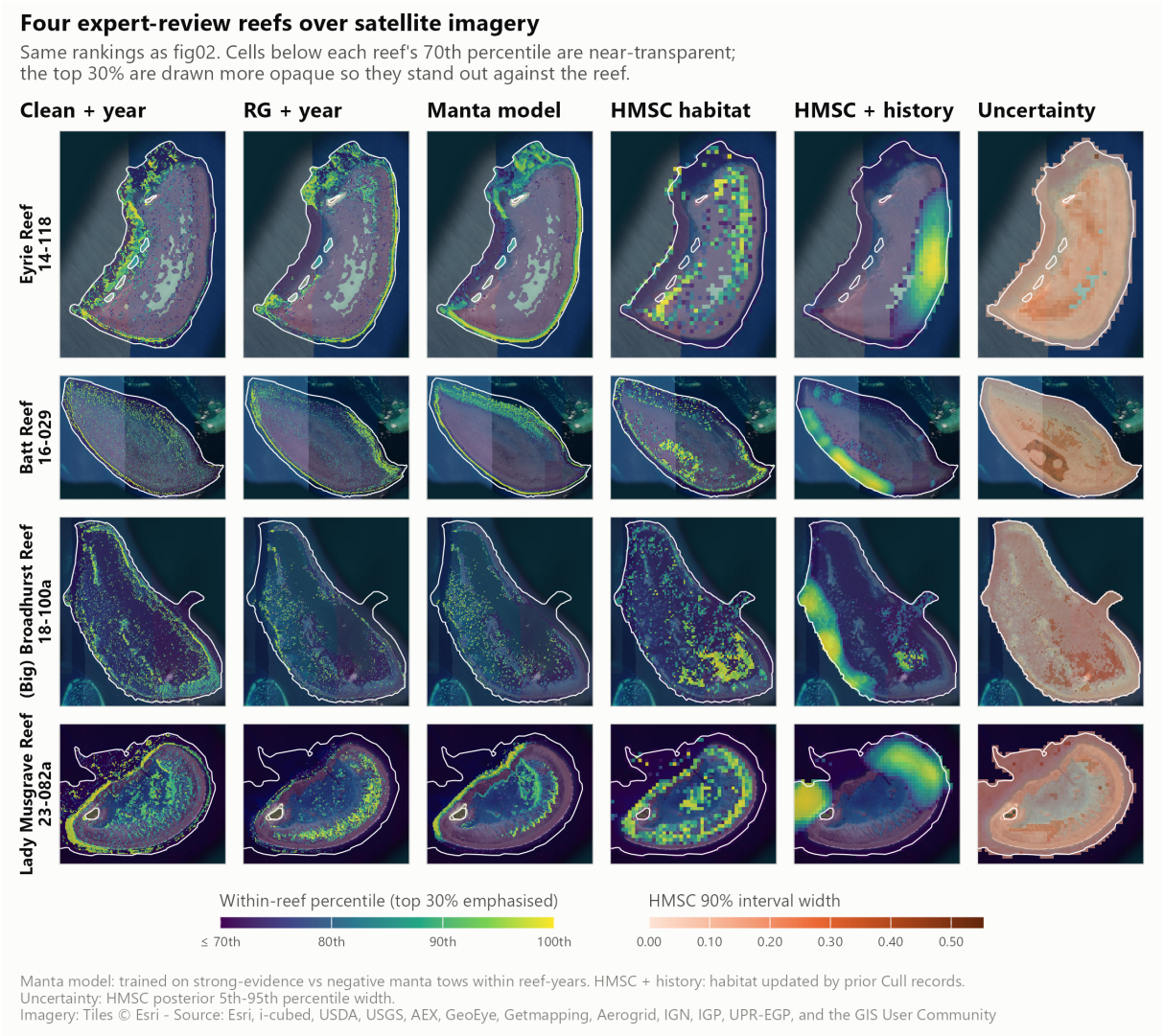


Figure 2: Within-reef percentile rank of each layer on four reefs (Eyrie, Batt, Broadhurst and Lady Musgrave) over satellite imagery. Columns: clean + year, RG + year, manta model, joint-model habitat, joint model updated with cull history, and the width of the joint model's 90% posterior interval. Cells below each reef's 70th percentile are drawn near-transparent. The layers agree on reef margins and outer slopes but differ in detail.

Table 2: Discrimination between strong-evidence and strictly negative manta tows within reef-years. Upper rows: independent check of the deployed habitat layers, none of which used manta tows. Lower rows: cull-trained and manta-trained models refitted on the same leave-reef-out folds and scored on held-out reefs. Lift is the share of strong-evidence tows in a reef-year's top 20%, divided by 20%. Brackets: 95% bootstrap intervals over reef-years.

Model	Tows	Within reef-year AUC	Top-20% lift
Clean + year (deployed layer)	117,318	0.516 (0.500–0.532)	1.11 (1.06–1.15)
RG + year (deployed layer)	117,318	0.576 (0.548–0.602)	1.23 (1.18–1.29)
HMSC joint (deployed layer)	117,318	0.567 (0.549–0.585)	1.15 (1.11–1.19)

Table 2: Discrimination between strong-evidence and strictly negative manta tows within reef-years. Upper rows: independent check of the deployed habitat layers, none of which used manta tows. Lower rows: cull-trained and manta-trained models refitted on the same leave-reef-out folds and scored on held-out reefs. Lift is the share of strong-evidence tows in a reef-year's top 20%, divided by 20%. Brackets: 95% bootstrap intervals over reef-years.

Model	Tows	Within reef-year AUC	Top-20% lift
Cull-trained (RG predictors + year)	198,852	0.547 (0.516–0.574)	1.15 (1.10–1.20)
Manta-trained (with year)	198,852	0.575 (0.547–0.599)	1.27 (1.21–1.33)
Manta-trained, within reef-year	198,852	0.612 (0.578–0.636)	1.33 (1.27–1.39)

Cull history forecasts problematic sites but encodes where crews dive

Problematic sites persisted between seasons. In the strictly forward test, the blend of habitat and cull history reached a within reef-year AUC of 0.681 (95% CI 0.640–0.718), against 0.569 (95% CI 0.529–0.607) for habitat alone (Table 3, Figure 3). Much of this skill reflects operator behavior: a score based only on how often crews had dived a site reached 0.636. The blend reached 0.716 where history existed nearby, but only 0.553 at sites never culled before. History lies within reach of 8.6% of 120 m reef cells. Reef-context predictors at 0.5–1 km, including windward position, distance to the reef edge and modeled currents, did not improve habitat forecasts robustly (0.582 vs 0.569, overlapping intervals).

Table 3: Forward validation of scores for problematic cull site-years: within reef-year AUC, test years 2022–2026 pooled. Every score uses only data from before the test year. History settings were fixed before validation. Sample sizes are site-years.

Score	All (95% CI) (n = 4,170)	Nearby history (n = 3,212)	Little or none (n = 958)	New sites (n = 1,664)
Habitat (joint model refitted before test year)	0.569 (0.529–0.607)	0.584	0.554	0.524
Habitat + reef context	0.582 (0.545–0.620)	0.600	0.520	0.514
Dive effort only	0.636 (0.602–0.664)	0.656	0.491	0.513

Table 3: Forward validation of scores for problematic cull site-years: within reef-year AUC, test years 2022–2026 pooled. Every score uses only data from before the test year. History settings were fixed before validation. Sample sizes are site-years.

Score	All (95% CI) (n = 4,170)	Nearby history (n = 3,212)	Little or none (n = 958)	New sites (n = 1,664)
History kernel	0.654 (0.601–0.700)	0.705	0.465	0.476
Habitat + history	0.681 (0.640–0.718)	0.716	0.540	0.553

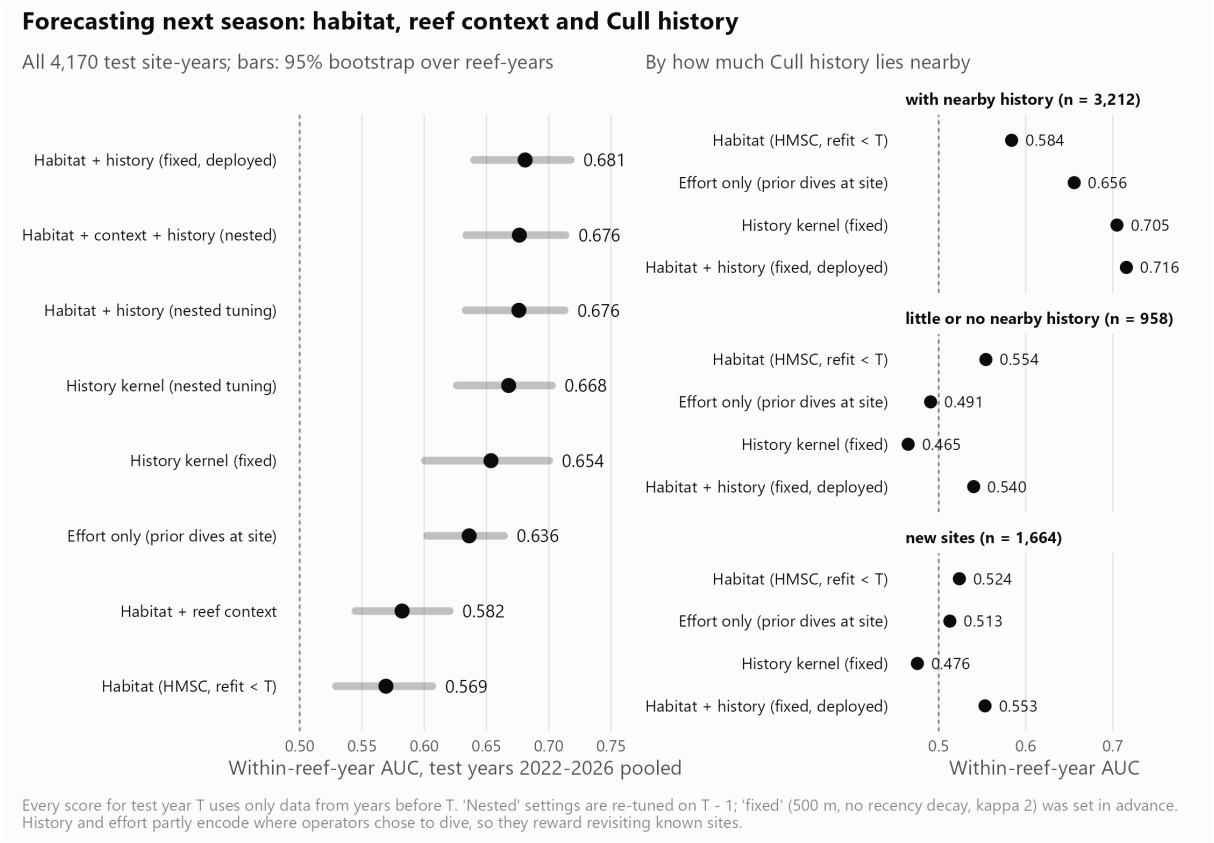


Figure 3: Forward validation of habitat, reef context and cull-history scores. Left: all test site-years, with 95% bootstrap intervals over reef-years. Right: the same scores split by how much cull history lies within reach of the site. History carries the forecast where it exists and adds nothing where it does not.

Evenly spaced manta tows retain more aggregations than model-ranked tows

At every level of effort, systematic thinning found the most aggregations, random thinning fewer, and model-ranked thinning the fewest (Table 4, Figure 4). With 60% of tows kept, systematic thinning found 96.6% of 1,900 aggregations in 1,082 surveys and random thinning 94.0%. Thinning by the held-out cull-trained layer found 88.1% (95% CI 86.6%–89.5%). The largest cut that still found at least 95% of aggregations was 40% of tows for systematic thinning (40% with bootstrap assurance), 30% for random thinning and 20% for the layer. Results for 2022–2026 alone were the same.

The manta model, which ranked tows best, found 3 percentage points more aggregations than the cull-trained layer at 60% kept (95% CI +1 to +5; Figure 5). This was still below both random (94.0%) and systematic thinning (96.5%). Model-ranked thinning did slightly better than random for isolated single strong tows (64% vs 61% at 60% kept), but these are not aggregations. Scoring tows with the deployed, in-sample layer overstated aggregations found by 2 percentage points at 60% kept.

Table 4: Share of COTS aggregations (two or more strong-evidence tows within 500 m) still found when manta tows are cut within each survey, all years. The model is the cull-trained RG + year layer scored leave-reef-out; 'deployed' is the same layer scored in-sample. Brackets: 95% bootstrap interval over surveys.

Tows kept	Model-ranked (held out)	Model-ranked (deployed)	Random	Systematic
80%	96.2% (95.3%–97.2%)	96.9%	98.8%	99.7%
70%	93.0% (91.8%–94.2%)	93.7%	97.0%	98.7%
60%	88.1% (86.6%–89.5%)	90.4%	94.0%	96.6%
50%	81.8% (80.0%–83.4%)	85.2%	89.7%	93.0%

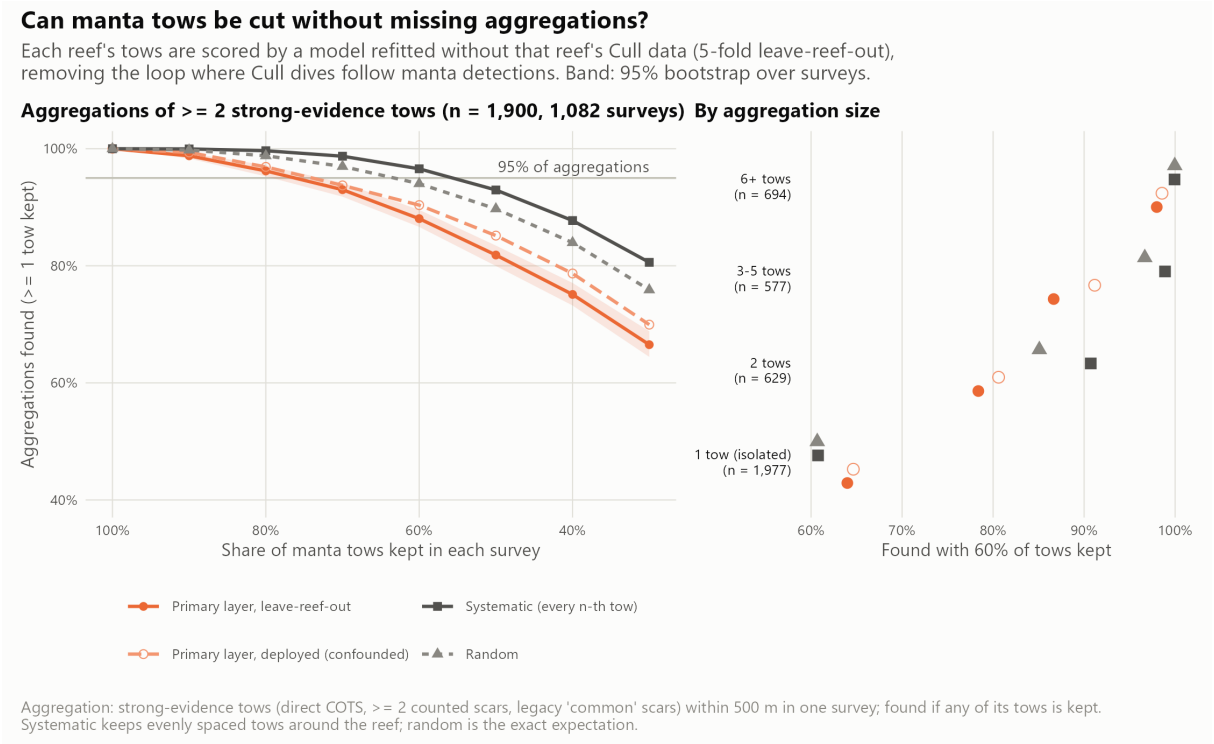


Figure 4: Manta tows kept versus COTS aggregations found. Left: aggregations of two or more strong-evidence tows found as tows are cut, for the cull-trained layer (held out and deployed), random and systematic thinning; bands are 95% bootstrap intervals over surveys. Right: share found at 60% of tows kept, by aggregation size. Evenly spaced thinning loses the fewest aggregations.

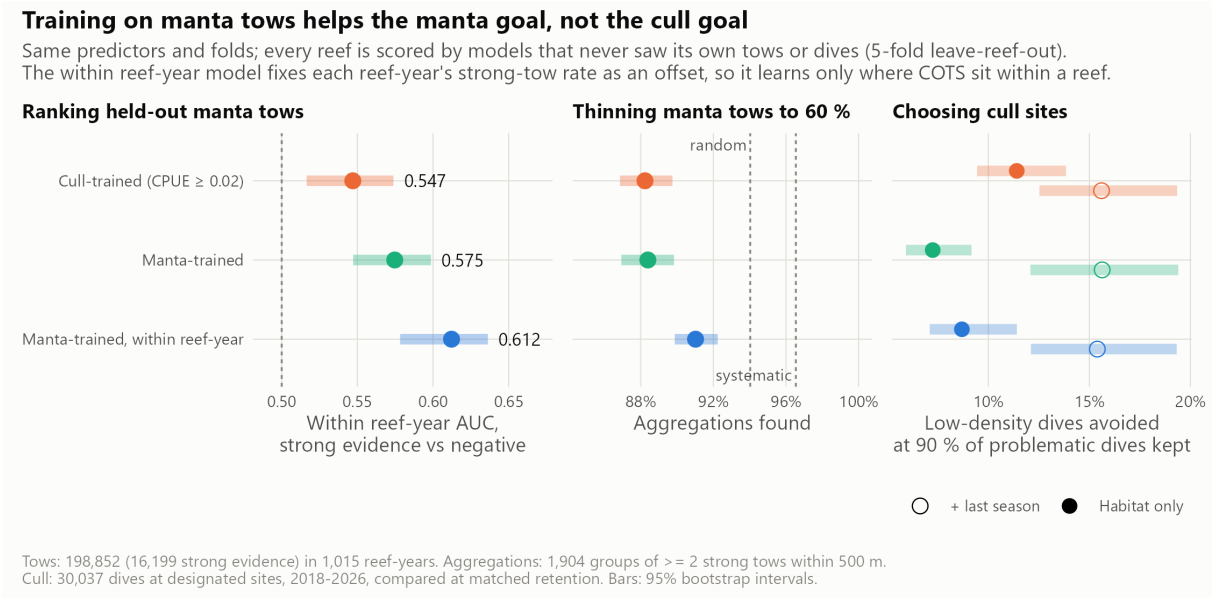


Figure 5: Cull-trained and manta-trained habitat models on both operational goals. Left: within reef-year AUC on held-out manta tows. Center: aggregations found with 60% of tows kept, with random and systematic thinning for reference. Right: low-density cull dives avoided at 90% of problematic dives retained, for the habitat model alone and with last season's evidence. Bars are 95% bootstrap intervals. Training on manta tows improves tow ranking but not cull-site selection.

Last season's evidence, not habitat, reduces the cull-site search

Habitat alone. Ranking all designated sites on a reef by the held-out habitat layer retained problematic dives only slightly faster than low-density or ghost dives (Figure 6). Searching the top 70% of sites retained 77.2% (95% CI 72.9%–80.8%) of problematic dives. Those sites accounted for 75.6% of all dives, which is what a ranking with no information would retain. Within reef-years, AUC was 0.57 for problematic versus low-density dives and 0.60 versus ghost dives. In-sample, the same layer appeared stronger (0.61 and 0.66). Training the habitat model on manta tows did not help: with survey year as a predictor, the manta model ranked cull dives at chance (AUC 0.48), because survey year took 70% of its fitted gain. The within reef-year version matched the cull-trained model (0.566 vs 0.567).

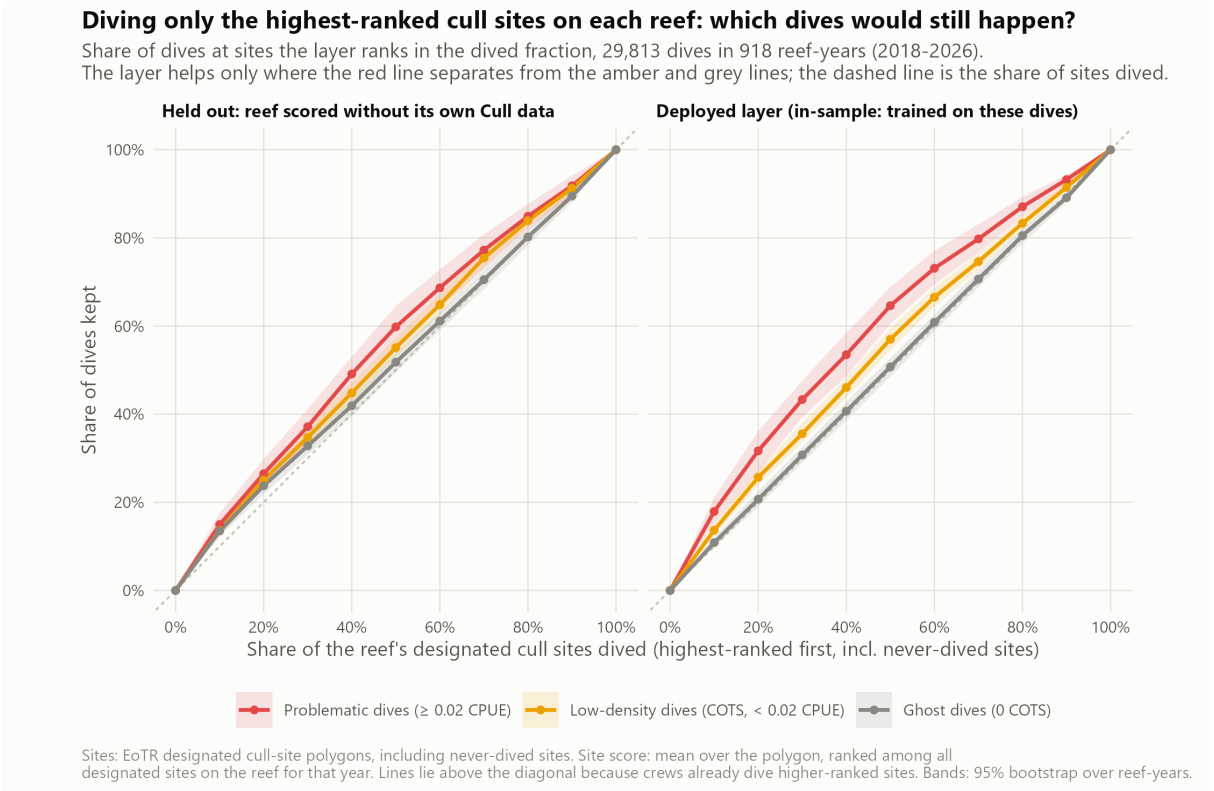


Figure 6: Share of cull dives that would still occur if only the highest-ranked designated cull sites on each reef were dived, by dive outcome (2018–2026). Left: held-out habitat ranking; right: deployed layer scored in-sample. The dashed diagonal is the share of sites dived. Bands are 95% bootstrap intervals over reef-years. On held-out data, problematic, low-density and ghost dives fall away at almost the same rate.

Adding last season's evidence. Evidence from the previous season reduced the search substantially (Table 5, Figure 7). Over 2018–2026 (30,037 dives in 926 reef-years), with 90% of problematic dives retained, the habitat + evidence ranking reduced designated sites searched from 88% to 82%. It raised low-density dives avoided from 10% to 16%, and ghost dives avoided from 13% to 21%. Every paired interval excluded zero. Gains were larger in 2022–2026. At 80% retention in that period, sites searched fell from 74% to 61%.

The evidence did most of the work:

- A ranking from evidence alone performed almost as well as habitat + evidence.
- Habitat contributed by ordering sites without recent evidence. On reefs not visited the previous season (3,221 dives), no ranking discriminated well (AUC about 0.53).
- Among dives actually made, discrimination between problematic and low-density dives rose only from 0.57 to 0.59. The reduction in search came mostly from ranking quiet and never-dived sites lower.

The fitted model raised a site's rank where strong manta tows or problematic dives had been recorded nearby. It lowered the rank sharply where the site had been dived the season before with nothing problematic. A site that had itself been problematic ranked slightly lower than its neighborhood suggested, consistent with culling having reduced it.

Table 5: Habitat-only ranking → habitat plus previous-season evidence, compared at matched retention of problematic dives (held out, leave-reef-out). Brackets: 95% paired bootstrap interval of the difference in percentage points.

Period	Problematic dives kept	Designated sites searched	Low-density dives avoided	Ghost dives avoided
2018–2026	90%	88% → 82% (-11 to -1)	10% → 16% (+2 to +10)	13% → 21% (+4 to +13)
2018–2026	80%	73% → 67% (-11 to -3)	22% → 27% (+2 to +10)	26% → 36% (+5 to +14)
2022–2026	90%	88% → 78% (-14 to -4)	10% → 19% (+6 to +13)	12% → 25% (+8 to +17)
2022–2026	80%	74% → 61% (-22 to -4)	20% → 32% (+6 to +20)	25% → 41% (+8 to +24)

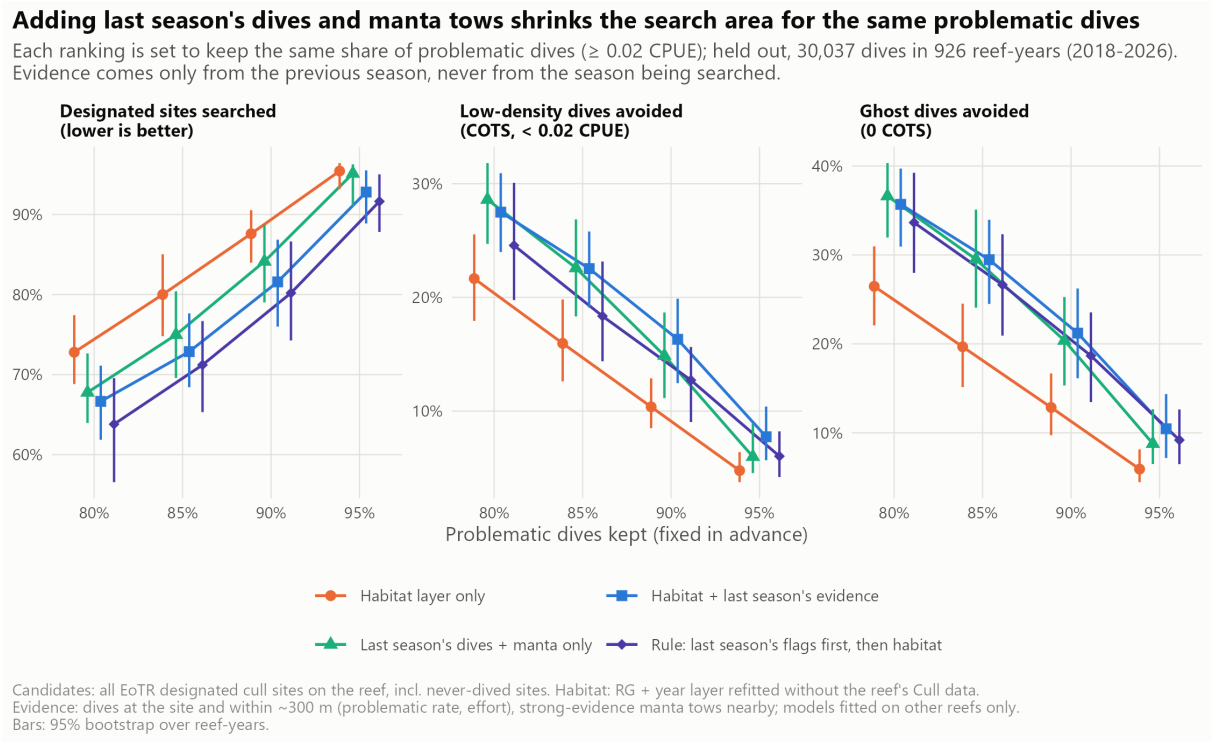


Figure 7: Habitat-only and evidence-informed rankings of designated cull sites compared at the same share of problematic dives retained (80–95%). Panels: designated sites searched (lower is better), and low-density and ghost dives avoided. Evidence comes from the previous season only. Bars are 95% bootstrap intervals over reef-years (2018–2026).

The gain holds forward in time

Fitted only on earlier seasons, the habitat + evidence model avoided more low-density dives than habitat alone in every season from 2022, and searched fewer sites in every season from 2023 (Table 6). It did not help in 2019–2021, when fewer earlier seasons were available for training. Pooled over 2019–2026 (28,431 dives in 861 reef-years), with 90% of problematic dives retained, sites searched fell from 86% to 81%, and low-density dives avoided rose from 11% to 17%. Training on all earlier seasons matched the last five (81% of sites) and outperformed the last three (86%). Adding the manta model to the habitat term made no difference (80%).

Table 6: Season-by-season forward test of the cull-site ranking with 90% of problematic dives retained. Each season is ranked by a model fitted only on earlier seasons. Values: habitat alone → habitat + last season's evidence. The 2026 season contains January–May only.

Test season	Dives	Sites searched	Low-density dives avoided	AUC, problematic vs low-density
2019	4,380	88% → 85%	11% → 14%	0.59 → 0.50

Table 6: Season-by-season forward test of the cull-site ranking with 90% of problematic dives retained. Each season is ranked by a model fitted only on earlier seasons. Values: habitat alone → habitat + last season's evidence. The 2026 season contains January–May only.

Test season	Dives	Sites searched	Low-density dives avoided	AUC, problematic vs low-density
2020	3,311	75% → 73%	23% → 21%	0.61 → 0.59
2021	3,769	78% → 80%	19% → 17%	0.54 → 0.56
2022	4,109	80% → 81%	13% → 15%	0.55 → 0.61
2023	4,119	89% → 74%	10% → 21%	0.59 → 0.61
2024	3,910	88% → 84%	7% → 15%	0.54 → 0.61
2025	3,676	88% → 84%	11% → 18%	0.52 → 0.60
2026 (Jan–May)	1,157	96% → 76%	5% → 25%	0.53 → 0.71
2019–2026	28,431	86% → 81%	11% → 17%	0.57 → 0.58

Deployed product for the 2027 season

The deployed ranking covers all 15,884 designated sites on 576 reefs (Figure 8). In the forward test, searching through classes A to C (the top 80% of a reef's sites) retained 89% of problematic dives while avoiding 18% of low-density and 22% of ghost dives. Class A alone (top 20%) retained 33% of problematic dives.

Evidence drives the top of the ranking. 678 sites had a problematic dive in the evidence window, and 809 had strong manta evidence nearby. 293 were dived without a problematic result. However, 10,654 sites (67%) lie on reefs with no dives or tows in the evidence window, so they are ranked by habitat alone.

The exploration layer contains 3,807 areas on 488 reefs (median 8 ha, median 170 m from a cull site). Habitat models still ranked tows on new ground, defined as no tow within 300 m in the previous five seasons. There the manta model reached a within reef-year AUC of 0.619 (95% CI 0.587–0.651), against 0.56 for the cull-trained layer. Strong evidence was more common on new ground than on repeat ground (13.6% vs 5.8% of tows).

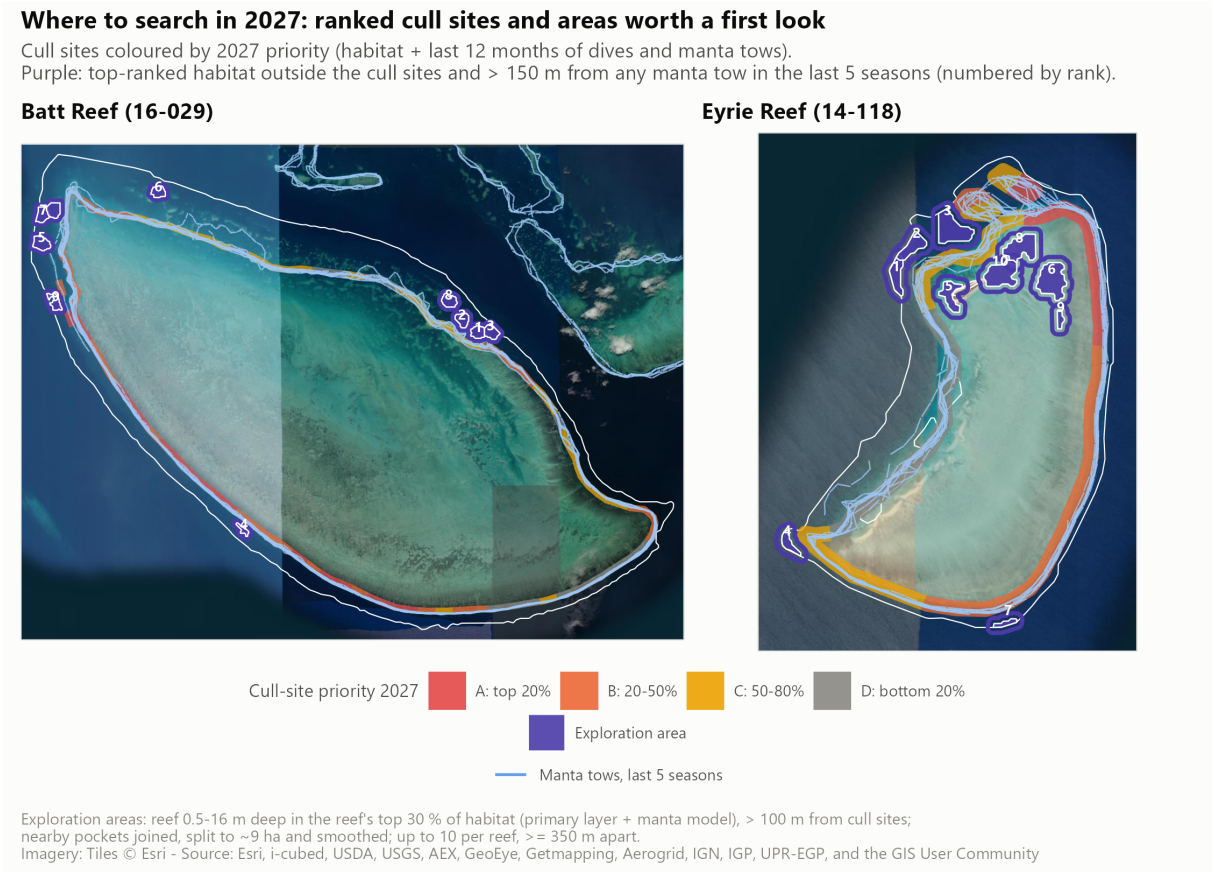


Figure 8: Deployed 2027 products on Batt Reef and Eyrie Reef: cull-site priority (A red to D grey), manta tows of the last five seasons (blue) and exploration areas (purple, numbered by rank) over satellite imagery. On these well-surveyed reefs, exploration areas fill gaps at reef edges and ends and, at Eyrie, part of the back reef that no tow has crossed in five seasons.

Metacommunity calibration and historical culling

Three configurations were retained for further testing: W, the working configuration; C3a, with a higher initial preferred-coral share at Lizard Island; and C6, with lower chronic background coral mortality. Across 5 ecological–connectivity seed pairs, each matched 7 of 7 observed historical peaks under score v2. Outbreak heights and troughs still differed (Figure 9; Table 7). C3a produced a first Lizard peak of 56.1 ± 6.5 adults ha^{-1}, compared with 29.0 ± 2.3 for W, but retained more COTS between outbreaks. C6 reduced trough densities while worsening the coral fit. The reported SDs describe variation across seeds, not uncertainty in biological parameters or observation conversion.

Passing the qualitative gate depended on the assumed density-dependent manta observation mapping. At $A½ = 30$ adults ha^{-1}, W passed in 5 of 5 seeds, C6 in 3 and C3a in 0. At $A½ = 10$, W passed in 0 seeds. These results were calibration diagnostics, conditional on a detection curve

410 that remains uncertain.

Table 7: Replicated metacommunity calibration on the four Lizard Island reefs (run 20261014T_48). Means ± SD across five recorded seed pairs; these are measures of stochastic variation, not confidence intervals. Lower COTS loss (score v2) and coral J indicate better calibration fit. Gate columns count seeds passing G2′ under the indicated density-dependent manta detection assumption, with A½ in adults per hectare. All candidates matched seven historical peaks in every seed at the linear observation mapping. No entry represents held-out validation.

Candidate	COTS loss v2	Coral J	Gate A½ = 10	Gate A½ = 30
W	6.79 ± 0.79	0.250 ± 0.002	0/5	5/5
C3a	6.72 ± 0.70	0.228 ± 0.002	0/5	0/5
C6	7.13 ± 0.85	0.279 ± 0.002	3/5	3/5

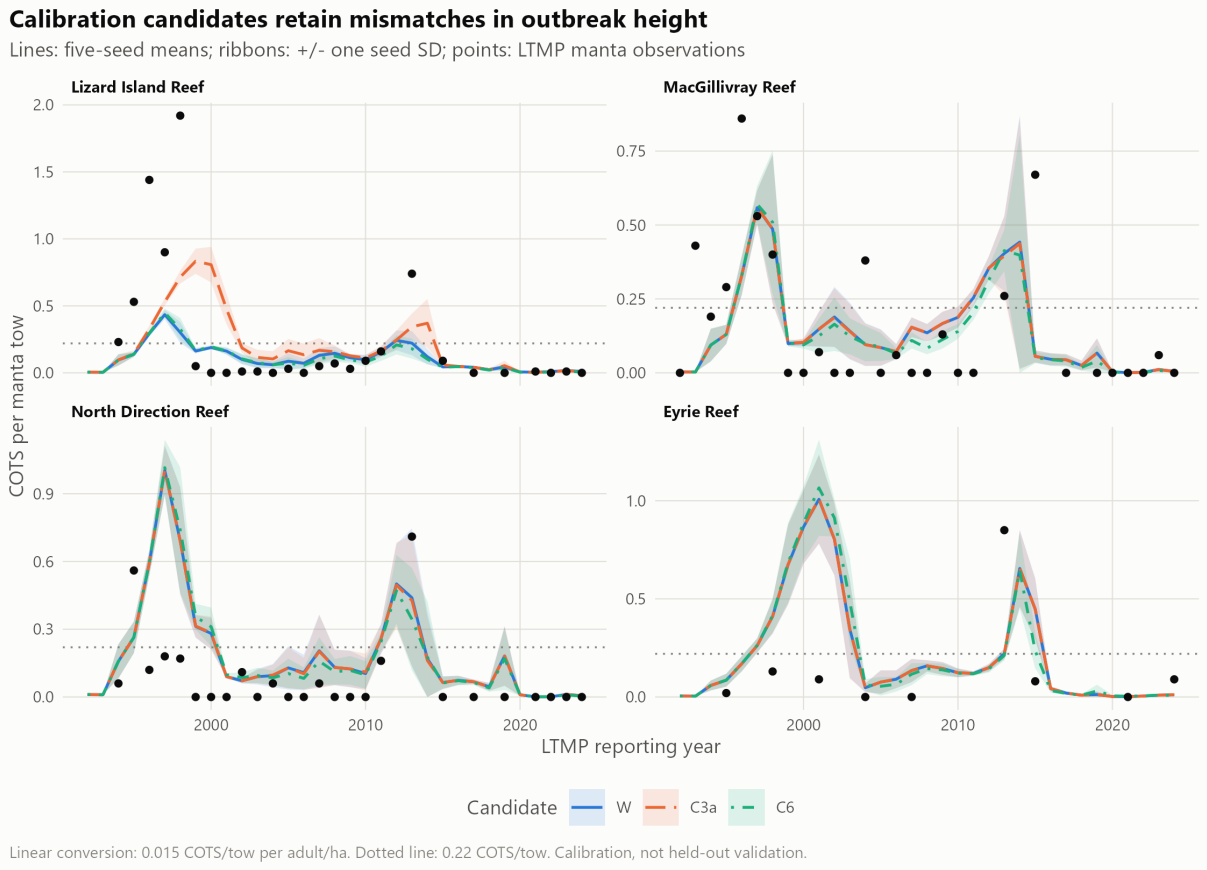


Figure 9: Simulated historical COTS trajectories on four calibration reefs. Lines are five-seed means for W, C3a and C6, ribbons are ± one SD across seeds, and points are LTMP manta observations. The linear conversion is 0.015 COTS per tow per adult ha^{-1}; the dotted line marks 0.22 COTS per tow. The successful peak counts do not imply accurate peak heights, troughs or independent predictive performance.

411 Prescribed historical culling exposed a more recent mismatch. With the primary 210 m spatial-

allocation kernel, the working model could not supply 88% of recorded removals at Eyrie during 2021–24, or 70% across the domain. Simulated populations were too small to support the observed catches. At North Direction, where removals could be applied, the simulated second-wave maximum fell from 0.82 to 0.55 COTS per tow. This single-seed test checked historical consistency and did not evaluate an adaptive control policy.

Diagnostic runs extending the domain to 2026 examined the missing recent build-up with culling off. Correcting the outbreak-boundary input with cull/SALAD evidence reduced score v3 from 15.61 to 12.63 under the earlier coral thermal settings. Revising those thermal settings gave scores of 11.34 with the original boundary and 11.34 with the corrected boundary. The best exploratory chlorophyll-survival variant under the corrected boundary scored 10.99. Score v3 incorporated a diver-anchored recent-density term and cannot be compared directly with score v2. These comparisons used one seed pair, and evidence used to correct the boundary also entered its assessment. They therefore identify candidate explanations for further testing. No configuration has completed held-out and regional validation, and benefits of primary-outbreak suppression or starting and stopping rules have not yet been estimated.

Discussion

Combining recent COTS observations with habitat information offers a practical way to improve daily search planning, while the metacommunity model provides a framework for evaluating control over longer periods and across connected reefs. The two strands have reached different stages. The search ranking improved retrospective retention of problematic dives for a given search effort, with recent observations carrying most of the information and habitat ordering locations where evidence was sparse. The population model reproduced important features of historical outbreaks but exposed unresolved differences in recent populations and coral dynamics. Its intended role is to assess the culling program's impact since 2012 and project how suppression of the fifth primary outbreak wave could affect future coral cover.

Combining habitat and recent observations for daily operations

The benefit of the site ranking comes from combining information with complementary coverage. Habitat layers describe reef settings across places with very different histories of observation. Recent dives and manta tows provide more direct evidence of current COTS populations,

but only where crews have searched. Their combination can improve search order even when habitat alone has weak discrimination. This is a useful distinction for deployment: broad habitat coverage provides a starting point, while new observations allow priorities to change as an outbreak develops and control proceeds.

The ranking also captures aspects of operator behavior. Crews select sites using previous observations and their knowledge of the reef, so a model trained on their records partly formalizes those choices. This is consistent with the sampling problems identified in ecological models fitted to preferentially collected observations (Phillips et al., 2009; Diggle et al., 2010). The retrospective gains therefore do not establish how much the ranking adds to experienced crew planning, nor how well it identifies problematic densities at never-dived sites. A prospective trial should compare its recommendations with current practice while recording search effort and outcomes.

Exploration areas can help broaden that evidence base. Historical tows entering previously unsurveyed ground recorded strong COTS evidence more frequently than repeat tows, although outbreak stage may contribute to this difference. That finding does not establish the yield of the proposed exploration polygons. Their value should be tested by recording both detections and nil results under a design that distinguishes new ground from repeat visits. Otherwise, a ranking learned from previous effort may progressively reinforce the existing search footprint.

Maintaining spatial coverage in manta surveillance

The manta results show why retaining high-scoring individual observations is not equivalent to retaining detections of whole aggregations. Model-ranked thinning concentrated observations in favored areas, leaving aggregations elsewhere vulnerable to being missed. Evenly distributed thinning maintained better coverage of the aggregations present in the observed surveys. Habitat can consequently be used to propose additional searches without making it the basis for removing surveillance from lower-ranked stretches.

This interpretation applies to retrospective thinning of the available survey footprints. Tow sequence was inferred from position around the reef, and the performance of newly designed routes remains untested. The evidence supports spatial coverage as a priority when effort is reduced within comparable perimeter surveys; it does not identify a universally optimal survey design. Operational trials should record actual tow order and routes so that future comparisons

can evaluate the coverage crews deliver.

Estimating the program's impact since 2012

The metacommunity model is intended to estimate how the culling program has altered COTS abundance and coral trajectories from 2012 to the present. Observed removals establish how many COTS crews killed. Estimating the broader impact requires a counterfactual: how populations, reproduction and coral consumption would have developed without those removals. A coupled COTS–coral model can represent those pathways and the effects carried by larvae to connected reefs, extending the management questions addressed by earlier population and ecosystem models (Morello et al., 2014; Matthews et al., 2020b; Condie et al., 2021; Bozec et al., 2022).

The proposed assessment would pair simulations with recorded removals against simulations omitting program removals from 2012 onward, starting from the same pre-control states and receiving matched environmental forcing and random seeds. Reproduction, dispersal and feeding would evolve within each trajectory. Differences in cumulative coral consumption, coral cover and recovery, alongside COTS abundance and larval export, would quantify the modeled consequences of control through the latest season supported by the records. This would estimate effects within the represented region and processes, rather than infer them directly from the number of animals killed.

The historical-culling diagnostic identified a prerequisite for that assessment: simulated populations must be large enough to support the observed catches. The current model could not supply much of the recorded recent catch on some reefs. Treating that discrepancy as uncertainty in control efficacy would confuse a population-model failure with an intervention outcome. The recent build-up must first be represented credibly, with the fit assessed against diver observations as well as manta records. Reproducing historical peak timing alone does not ensure that the abundance available for culling or its consumption of coral is correct.

The calibration comparisons also show how different ecological explanations can compensate for one another. Starting with more preferred coral at Lizard Island produced a larger first outbreak but left more COTS between outbreaks. Reducing background coral mortality lowered those intervening populations while worsening the coral fit. Assumptions about how often manta observers detect COTS at low density changed whether the model met the between-outbreak cri-

teria. These alternatives should remain visible in estimates of program impact; selecting a single configuration by its aggregate fit would conceal uncertainty in the mechanisms producing the estimated benefit.

Historical attribution further depends on the external larval boundary. Holding larval supply from outside the modeled region fixed estimates the effect of control within that region under the same external pressure. Estimating the wider program's contribution requires control on external source reefs to be represented too. In particular, outbreak inputs corrected using observations made under control cannot automatically be treated as the boundary for a world without control. The counterfactual should preserve this distinction and report how alternative boundary assumptions change the estimated coral benefit.

Projecting suppression of the fifth primary outbreak wave

The model's forward-looking purpose is to assess whether suppressing the fifth primary outbreak wave can protect coral in the northern initiation region and reduce subsequent larval pressure on connected reefs. The proposed comparisons would include continuation of current control, earlier or more sustained suppression, delayed intervention, and alternative rules for beginning or ending culling. A no-control trajectory would provide context, while comparisons among active strategies should use matched resource budgets. This separates benefits arising from the timing and location of control from those arising simply from greater effort.

The projections should track both population and coral outcomes. Suppression may change the duration of high COTS densities, the larvae exported from treated reefs and the timing of outbreaks farther downstream. The resulting coral consequences depend on the coral available for feeding, its recovery and future disturbances. Coral protected from COTS may later be lost to bleaching or cyclones, while maintaining coral through one outbreak may alter the food available to subsequent populations. Paired simulations under the same environmental sequences would allow these effects to be followed through time without assuming that a reduction in COTS translates proportionally into lasting coral protection.

These projections require further development beyond prescribing historical catches. The model must connect feasible vessel and diver effort to detection and removals as COTS become scarce, and apply explicit starting and stopping rules. The daily site ranking could eventually help distribute search effort within those scenarios, but that coupling has not been validated. Projection

horizons, effort budgets and climate scenarios also need to be agreed with the program. Their comparison should retain uncertainty in the present outbreak state, population processes, observation methods and larval transport, rather than rely on a single best-fitting historical trajectory.

Evidence needed for operational and strategic decisions

The two strands require different evidence before their benefits can be claimed. The site ranking can now be trialed against crew planning, recording priorities before visits, actual effort, departures from the suggested order and outcomes. The metacommunity model requires independent checks of populations and coral trajectories, including held-out reefs and years, regional transfer and sensitivity to biological and boundary assumptions. Its latest extended runs used one seed pair and shared cull and SALAD evidence between corrected inputs and assessment. Their improved fit is a diagnostic result, not independent evidence of predictive skill or control benefit.

For the search ranking, spatial and temporal tests should also remain distinct. Habitat scores were held out by reef, and the season-by-season evidence regression was trained on preceding seasons. The habitat fits in that comparison were not themselves restricted to earlier years. A future prospective evaluation should reproduce the entire information available at the planning date. More generally, validation should match the spatial and temporal decision being supported (Roberts et al., 2017), and discrimination should be assessed within reefs rather than inferred from pooled differences among reefs and years (Lobo et al., 2008).

Baselines are essential to both strands, but they answer different questions. A no-information search ranking shows how much a planning tool adds at the same effort; a comparison with crew practice shows whether it improves an existing decision process. A historical no-control simulation estimates the modeled contribution of the program, whereas future comparisons at matched budgets test how its resources could be allocated more effectively. Keeping these questions separate prevents either effort reduction or a better calibration fit from being credited as a demonstrated management benefit.

Deployment and monitoring

The 2027 product should guide search order, with opportunities to revisit lower-ranked sites and explore beyond recent effort. Monitoring should record the priority available before each visit,

actual searches and effort, and problematic, low-density and ghost dives. Exploration outcomes should be assessed separately from repeat ground, allowing for outbreak stage. Updating the ranking and repeating its evaluation each season would show whether gains persist as crews, outbreak conditions and the information base change. [TO CONFIRM: trial design, monitoring owner and reporting route agreed with GBRMPA and vessel operators.]

For the strategic assessment, the immediate priorities are to reconcile recent diver and manta evidence, represent the observed populations exposed to culling, and validate coral and COTS trajectories independently. The subsequent paired historical assessment would estimate the program's effects from 2012 to the present. Forward scenarios would then examine fifth-wave suppression and the consequences for local and downstream coral. [TO CONFIRM: historical endpoint, projection horizon, intervention budgets and climate scenarios to be compared.]

Conclusion

Recent observations combined with habitat provide an actionable basis for testing more efficient daily searches. The metacommunity model addresses the longer-term question of how those removals change outbreaks and coral across connected reefs. Its planned historical and future comparisons can make the consequences of program decisions explicit, provided population, observation and coral uncertainties are carried into the assessment. The present work supports field testing of the search product and further validation of the strategic model; it does not yet establish the coral benefit of suppressing the fifth outbreak wave.

Data availability statement

COTS Control Program cull and manta records are held by the Great Barrier Reef Marine Park Authority and are available on request, subject to its data-sharing conditions. [TO CONFIRM: access statement and contact.] The habitat layers, 2027 site ranking, results tables and analysis code are available at [TO CONFIRM: repository and DOI].

Ethics statement

[TO CONFIRM: e.g. "Ethical review and approval was not required for this study because it used existing operational monitoring records and involved no new field sampling or animal

handling.”]

Author contributions

[TO CONFIRM: CRediT roles for each author.]

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

SM declared that they were an editorial board member of Frontiers, at the time of submission. This had no impact on the peer review process and the final decision. [TO CONFIRM: add the same declaration for any other author who is a topic editor of this Research Topic, e.g. TI.]

Generative AI statement

[TO CONFIRM: whether generative AI was used, and if so the tool, version and purpose.]

Supplementary material

[TO CONFIRM: final list.] Proposed supplementary files:

- Supplementary Table S1: delivered layers and products (file, resolution, contents).

- Supplementary Table S2: predictor definitions and sources.
- Supplementary Table S3: joint-model convergence and coverage by component.
- Supplementary Figure S1: predictor importance in the cull-trained models.
- Supplementary Figure S2: within-reef agreement between layers.
- Supplementary Figure S3: effort cut by the habitat layer versus at random (2022–2026).
- Supplementary Figure S4: example manta and dive plans on two reefs.

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